Prehistoric VR simulations as educational content : Prehistoric VR Timeline

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Traditional visual timelines such as banners, infographics, Static maps are currently used to educate learners from museum, exhibit visitors to students, but with the static, heavy, one-way learning has often resulted in low engagement and poor memory retention. This research proposes a Prehistoric VR Timeline that uses enhanced learning methods (VAK learning model, gamifications, storytelling) to improve the historical learning outcomes. In this research we are using immersive Virtual Reality (VR) environments, narrations, pre-questions, and interactive micro-activities to implement those learning methods, final system aims to foster curiosity, enhance focus, and strengthen long-term memory. The methodology compares whether those outcomes have improved compared to traditional methods. The prototype shows a potential in improving traditional learning mediums by combining learning methods and technology to fill the gap of traditional learning mediums.

Keywords: Virtual Reality, Educational Technology, Prehistoric Timeline, VAK Learning Model, Interactive Learning, Memory Retention

I. INTRODUCTION

Currently in museums, exhibits, schools learning prehistoric timelines relied on traditional learning mediums such as banners, charts. While structured, they are text heavy and one way learning mediums that less attention grabbing and memory retentive, with that most museum visitors and students struggle to retain information, leaving a gap of memory retention. The problem is to present prehistoric events in an engaging and memorable way.

Virtual reality (VR) is a promising technology that offers immersive, interactive and multi-sensory experiences for educational purposes. Unlike static timelines VR can take historical events into reality with realistic prehistoric environments, animals that are impossible to experience in real life. This can make learning history from abstract dates into interactive, story driven experience that promotes deeper connection between the learner and the content

This paper introduces the Prehistoric VR Timeline, built with effective learning methods providing 360° environments, narrated guidance, and micro-interactions to improve curiosity and memory retention, unlike traditional methods it encourages prediction, interaction that give participation in the learning process.

The study investigates whether a VR-based timeline can outperform traditional methods in engagement and knowledge retention. By comparing user outcomes, it aims to show how immersive technologies can transform history education into a more effective and lasting experience.

II. LITERATURE REVIEW

A. Identified Knowledge Gap

From the reviewed literature, we have two major gaps that have emerged. First, some existing solutions use two-dimensional (2D) web applications that don't support the immersive learning environments and Storytelling 192670that improve engagement, limiting the potentially improved effectiveness of the experience. Second, even though some existing solutions use VR technology they focus on just visualizing the content without letting the learner participating in the experience, and they did not have implemented any learning techniques that improve engagement and memory retention

B. Research Objectives

The primary objective of this research is to design, develop and test a VR based prehistoric timeline that improves user engagement and memory retention compared to traditional methods

When it comes to specific objectives we have to deliver engaging prehistoric content within immersive VR environments, implement narrations, pre questions and micro interactions, apply VAK model properly and test the prototype with users to compare its results with traditional learning methods

C. Research Questions

"Can we teach people about prehistoric timelines better using VR interactive timeline that includes narration, Pre questions, interactions and gamification?" This study investigates standard ways of teaching prehistoric subjects may be improved by using immersive VR experiences. Unlike static mediums, VR allows students to interact directly with the subject, sparking curiosity and enhancing attentiveness through involvement. The proposed VR timeline includes narrative for guided learning, pre-questions to activate past information, and micro-interactions that allow users to investigate Mesozoic ecological milestones. Gamification elements during the recap stage improve memory retention and create a sense of accomplishment. By combining these components, the study analyses if learners obtain a deeper comprehension, enhanced recall, and increased engagement compared to traditional ways. The findings will help to understand VR's potential as an innovative teaching tool in both formal and informal learning situations

D. Hypotheses

Following hypothesis is put forth in considering the gaps and goals that have been highlighted.
H1 - When compared to those who use traditional visual timelines, users of the VR timeline will show noticeably better memory recall and engagement.

III. RESEARCH GAP

Studies on teaching using timelines has improved from traditional methods to digital once. However ,each step of this evolution still leaves some gaps in effectiveness ,especially with regard of engagement ,interactivity and memory retention. This gap can be explained by the Blue area, red area, green area framework.

Blue Area

Existing studies show attempts to modernize timeline-based history education:

- ACM (2021**)** introduced a 2D browser-based interactive timeline that improves navigation of historical periods.
- ProQuest (2020) developed another 2D digital timeline for exploring historical events more flexibly than printed materials.
- Zhang et al. (2023) integrated SCORM courseware with 360° panoramic VR videos to increase visual immersion in history lessons.

These works contribute to improved accessibility, structured content delivery, and enhanced visual engagement.

Red Area

Despite the progress, notable gaps remain

- 2D Constraint (ACM 2021; ProQuest 2020) -
  Navigation is possible, but there is *no interaction* with timeline events, keeping users passive.
- Passive VR Experience (Zhang et al. 2023) -
  360° videos offer immersion but lack object interaction, exploration, or kinesthetic engagement.
- No Memory-Retention Techniques (All Three Studies) -
  None include pre-questions, prediction prompts, micro-interactions, or structured cognitive cues.
- Limited Learning Modalities -
  Approaches rely mostly on visual (and sometimes audio) input, with no support for kinesthetic learning aligned with VAK principles.

These limitations restrict engagement depth and reduce long-term recall.

Green Area

The identified gaps create opportunities to advance timeline-based learning through:

- Interactive VR Timelines -
  Enabling users to interact with events, objects, and environments rather than only viewing them.
- Integration of Retention-Focused Techniques -
  Applying pre-questions, narration, micro-activities, and spatial memory cues to improve recall.
- VAK-Aligned Multisensory Learning -
  Combining visual, auditory, and kinesthetic interactions within VR to support diverse learning preferences.
- Active, Story-Driven Exploration -
  Converting timelines into guided experiences where learners predict, explore, and participate, rather than passively observe.

This highlights the need for a VR system that unifies interactivity, multisensory engagement, and retention-oriented learning design.

IV. METHODOLOGY

A. Research Design

This study used a two-part research design that included (1) an experimental comparison of the Prehistoric VR Timeline with a conventional static timeline and (2) an assessment of the VR system's usability. To evaluate how the two teaching approaches differed in terms of learning outcomes, engagement, and retention, the experimental component used a pre-test/post-test control group model. Understanding the navigation experience, interaction clarity, and usability for novice VR learners were the main goals of the usability component. These elements worked together to allow for a thorough evaluation of the practical usability and educational impact.

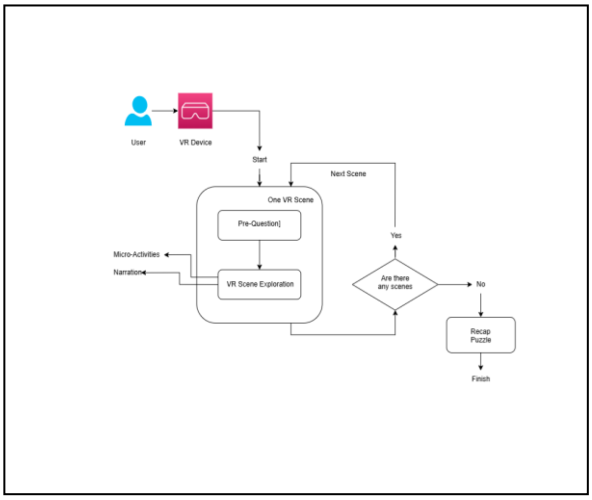

Figure 01: System Diagram

B. System Overview

A structured chronological sequence spanning the Triassic, Jurassic, and Cretaceous eras was used to create the Prehistoric VR Timeline in Unity. Pre-question prompts, immersive 360° environments, narrated guidance, micro-interactions like handling objects and animal behaviors, and a recap puzzle that serves to reinforce important ideas are all included in the experience. In standalone mode, the system is installed on the Meta Quest 3 headset. To guarantee content equivalency, a static timeline with the same prehistoric events, images, and descriptions was created for comparison. This made it possible to compare learning outcomes solely based on the type of instruction rather than variations in content.

C. Materials and Tools

Blender was used to model/ Modify the creature assets for the VR system, which was implemented using Unity 3D. Luvvoice and Pixabay was used to create the narration and get sound effects, and Unity was used to create the basic user interface. The Meta Quest 3, which ran without a PC to preserve accessibility, served as the hardware platform for VR testing.

The static timeline, which mirrored the order and factual content of the VR experience, was created as a high-resolution printed poster for the educational comparison. To standardize data collection, Google Forms was used to administer all tests and surveys.

D. Experimental Procedure

Two participant groups were exposed to identical prehistoric content as part of the educational comparison. To determine baseline knowledge, a pre-test was completed by both groups. While the VR Group experienced the VR system, which included interactive exploration, prediction prompts, and narration, the Static Group studied the printed timeline. Following instruction, a post-test measuring short-term comprehension was completed by both groups. This was followed by an engagement survey that evaluated focus, enjoyment, clarity, curiosity, and recommendation intention.

Five to seven days later, a delayed post-test was given to gauge retention. To lessen the effects of memorization, all tests included parallel question structures with different wording and randomized ordering.

Six to ten participants, roughly half of whom had never used virtual reality, participated in a usability evaluation in addition to the educational comparison. Every participant spoke their thoughts out loud as they finished the entire VR sequence. Navigation, instruction clarity, interaction attempts, and moments of confusion were all noted.

E. Data Analysis

Knowledge gain (Post-Test – Pre-Test) and retention percentage (Delayed Test ÷ Immediate Post-Test × 100) were calculated as part of the quantitative analysis for the educational comparison. To find variations in both short-term and long-term retention, group means for the two learning modalities were compared. Descriptive analysis was used to assess perceived clarity and engagement with the survey results.

Task completion rates, interaction success, time spent on each scene, and confusion point frequencies were recorded for the usability study. Overall usability was calculated by averaging ease-of-use ratings. To find reoccurring issues, interaction bottlenecks, or areas where the VR interface needs to be improved, observational notes were examined.

F. Ethical Considerations

Each process is compiled with recognized ethical guidelines for user-involved research. After being made aware of the study's objectives and procedures, participants provided their informed consent before it started. Participants were free to stop the VR exposure at any time to reduce the potential of motion discomfort. All responses were kept anonymous and utilized exclusively for research purposes. No personally identifiable information was collected, and participants were informed of their right to withdraw at any time without paying any fees.

V. RESULTS

The study's conclusions show that the Prehistoric VR Timeline clearly outperforms conventional static timelines in terms of user engagement and learning efficacy. Immersion and interaction in learning environments promote better comprehension and long-term memory consolidation, as evidenced by the VR group's noticeably higher knowledge gains and retention rates. VR participants outperformed static learners, who recorded lower gains (67.1%) and weaker retention (56.2%), with an average gain of 88.2% and a retention rate of 71.3%. H1 is directly supported by these findings, which show that VR learners exhibit notably superior memory recall compared to those who use conventional visual timelines.

H1 is further supported by the engagement data. When compared to the static timeline, participants gave the VR experience significantly higher ratings for enjoyment, focus, and clarity. These enhancements imply that immersion is essential for maintaining focus and minimizing cognitive exhaustion during the learning process. The hypothesis that VR users are more engaged than static-timeline learners is supported by the higher curiosity and recommendation scores among VR users, which also show that immersive environments significantly increase learner motivation.

The Prehistoric VR Timeline is usable even by beginner VR users, according to the usability results. Confusion points were low, task completion rates were consistently high, and ease-of-use ratings ranged from 4 to 5. These findings show that the system's interaction design is simple to use and has a low learning curve. The low frequency of motion discomfort lends even more credence to VR's usefulness for brief learning sessions. The system's suitability for implementation in educational settings was demonstrated by the participants' ability to move through scenes, initiate interactions, and finish the recap puzzle on their own.

When taken as a whole, the usability and educational results strongly support the updated hypothesis. VR proved that immersive learning environments can perform better than traditional static materials in both cognitive and experiential dimensions by improving memory recall and increasing engagement. The static timeline provided accurate information, but because it was passive, learners were less engaged, which led to poorer performance. Although more extensive content coverage and larger sample sizes are needed for future research, the current findings provide convincing preliminary evidence that VR timelines can significantly enhance the educational process.

ID	Group (Static/VR)	PreScore	PostNowScore	PostDay7Score	Gain (PostNow-Pre)	Retention (%)	Enjoyment	Focus	Clarity	Curiosity	Recommend
1	Static	3	5	5	66.70%	100%	2	2	3	2	5
2	Static	5	5	5	0%	100%	2	1	4	5	5
3	Static	3	3	4	0%	133.30%	3	1	4	4	7
4	Static	5	6	5	20%	83.30%	3	1	5	3	3
5	Static	5	7	6	40%	85.70%	1	1	3	2	4
6	Static	2	3	4	50%	133.30%	4	3	5	5	2
7	Static	1	0	1	-100 %	invalid	5	5	2	3	4
8	Static	4	3	4	-25 %	133.30%	1	2	6	2	7
9	Static	7	6	6	-14.3 %	100%	4	5	7	5	6
10	Static	2	4	2	100%	50%	3	4	3	6	3
11	Static	1	4	2	300%	50%	5	3	2	6	4
12	Static	1	6	3	500%	50%	2	4	6	5	1
13	Static	4	5	4	25%	80%	6	3	4	3	3
14	VR	5	8	8	60%	100%	6	6	7	7	9
15	VR	3	6	5	100%	83%	8	5	8	8	8
16	VR	2	6	6	200%	100%	6	7	8	6	8
17	VR	1	5	5	400%	100%	8	4	9	7	7
18	VR	8	9	9	12.50%	100%	9	8	7	5	6
19	VR	4	7	6	75%	85.70%	7	8	8	4	8
20	VR	5	7	5	40%	71.40%	10	7	9	8	5
21	VR	6	8	8	33.30%	100%	9	6	6	6	7
22	VR	2	4	4	100%	100%	8	9	7	7	5
23	VR	1	5	5	400%	100%	7	6	8	5	9
24	VR	2	6	4	200%	66.70%	10	8	4	9	9
25	VR	4	5	4	25%	80%	10	4	7	9	8
26	VR	1	8	7	700%	87.50%	9	7	5	8	7
					Avg Gain (%)(VR) = 88.2%	Avg Retention (%)(VR) = 71.3%	Avg Enjoyment(VR) = 4.7/5	Avg Focus(VR) = 3.9/5	Avg Clarity(VR) = 3.5/5	Avg Curiosity(VR) = 4.2/5	Avg Recommend(VR) = 4.1/5
					Avg Gain (%)(Static) = 67.1%	Avg Retention (%)(Static) = 56.2%	Avg Enjoyment(Static) = 2.1/5	Avg Focus(Static) = 1,5/5	Avg Clarity(Static) = 1.9/5	Avg Curiosity(Static) = 3.4/5	Avg Recommend(Static) = 2.9/5

Figure 02: Test Results - Educational impact: static vs VR

ID	VR Experience Level (Y/N)	Time (min)	Assists	Errors	Confusions	Completed Tasks	Avg Ease Score (1-5)	Motion Discomfort (Y/N)
1	N	2	1	4	0	1,2	4	N
2	N	2	1	2	0	2,3	5	Y
3	N	2	2	2	1	3,4	5	N
4	Y	2	1	3	0	4,5	5	N
5	N	2	1	2	0	5,6	5	N
6	Y	2	1	5	2	4,6	4	N

Figure 03: Test Result - Usability Testing

VI. DISCUSSION

The results of this study show that the Prehistoric VR Timeline clearly outperforms conventional static timelines in terms of user experience and learning efficacy. Immersive and interactive learning environments can improve both short-term comprehension and long-term memory consolidation, as evidenced by the VR group's noticeably higher knowledge gains and retention rates. VR participants outperformed static learners, who recorded lower gains (67.1%) and weaker retention (56.2%), with an average gain of 88.2% and a retention rate of 71.3%. The assertion that active engagement strategies, like narratives, prediction-based pre-questions, and micro-interactions, serve as powerful cognitive anchors and increase the memorability of prehistoric content is empirically supported by these findings.

The engagement statistics support VR's educational advantages even more. When compared to the static timeline, participants gave the VR experience significantly higher ratings for enjoyment, focus, and clarity, indicating that immersion is crucial for maintaining attention and preventing cognitive fatigue. In line with educational psychology models that connect engagement to better learning outcomes, VR users' higher curiosity and recommendation scores also imply that immersive environments may increase users' motivation to learn. The lower retention performance of static learners, on the other hand, was probably caused by their frequent reports of having trouble focusing.

The Prehistoric VR Timeline is usable even by new VR users, according to the usability results. Confusion points were minimal, task completion rates were consistently high, and ease-of-use ratings ranged from 4 to 5. These findings show that the system's interaction design is simple to use and has a low learning curve. Additionally, the low rate of motion discomfort suggests that the VR experience is suitable for brief learning sessions, which is a crucial factor to take into account when implementing it in educational settings like museums or classrooms. The system's suitability for practical application is demonstrated by participants' ability to move through scenes, initiate interactions, and finish the recap puzzle without outside help.

VII. CONCLUSION

In contrast to conventional static timelines, this study investigated the Prehistoric VR Timeline as a learning tool. The VR experience greatly enhanced memory recall and engagement, according to the results, with users showing increased learning gains, improved retention after a week, and increased focus and enjoyment. These results unequivocally support the hypothesis H1, demonstrating that VR learners perform better in terms of engagement and recall than users of static timelines.

The VR system was also very easy to use. The majority of participants, including those who had never used virtual reality before, navigated scenes, initiated interactions, and finished tasks with little help. The system is appropriate for use in educational settings, as evidenced by the high ease-of-use ratings and low motion discomfort reports.

Despite the prototype's limited coverage of three prehistoric eras and the small sample size, the study offers compelling evidence that immersive virtual reality can improve history education. Future research will concentrate on testing with larger sample sizes and adding more complex content. All things considered, the Prehistoric VR Timeline shows that, in contrast to conventional techniques, interactive, multisensory VR can produce more captivating and unforgettable learning experiences.

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